# P180 | Axial-piston pump

Compact, water-lubricated axial-piston technology for a defined operating duty.

P180 supplies pressurised water to hydraulic actuators or a process. Select delivered flow at the working pressure, then match displacement, speed, drive power and inlet supply.

| Published parameter | P180 range rating |
|---|---|
| Maximum displacement (cm³/rev) | 225.0 |
| Minimum displacement (cm³/rev) | 135.0 |
| Maximum speed, unboosted (rpm) | Not listed |
| Maximum speed, boosted (rpm) | 2000 |
| Maximum input power (kW) | 114 |
| Maximum water flow (L/min) | 430 |
| Maximum continuous pressure (bar) | 160 |
| Mass (kg) | 82 |
| Maximum temperature (°C) | 50 |
| Minimum temperature (°C) | 2 |

Nominal ordering displacements: 134 / 180 / 225 cm³/rev. Nominal options and range-table limits are different source entries; use the selected configuration.

## Design & environmental advantages

Multiple pistons smooth the delivery of power. Water lubricates the working surfaces, removing hydraulic oil from this circuit. Standard stainless construction and selectable internal materials support fluid-specific selection.

## Efficiency & selection

Match component size to useful work. Read the supplied efficiency curve at pressure and speed. Avoid sustained relief flow and excessive throttling; these convert useful hydraulic power into heat.

## Installation and procurement

Specify flow, pressure, speed, fluid chemistry, inlet supply, rotation, seal type, shaft coupling and mounting. No induced radial or axial shaft load. Nominal 10 µm / 25 µm absolute filtration, beta10 ≥ 75, is listed in the pump catalogue. Variable versions are listed for P60 and P180 only.

The range table lists a minimum of 135 cm³/rev; nominal ordering and some curve labels use 134. Match the quotation to its approved displacement and drawing.



Family photograph; configuration varies.

## Connect the selected design

Two-port pumps need flooded, positive inlet pressure and cannot lift water. Three-port boosted units separate the feed and case drain. Specify that arrangement in the order.

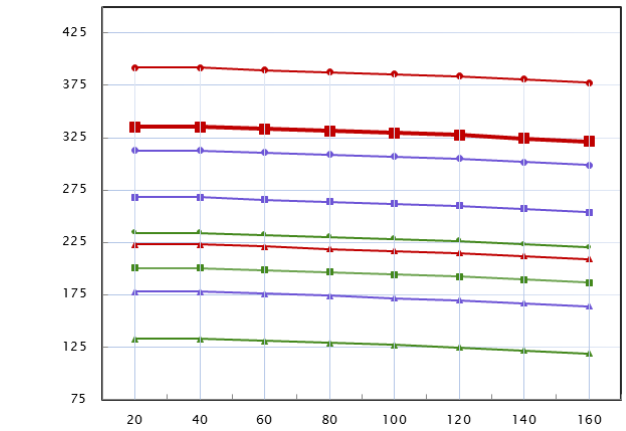
Reference conversions: 160 bar = 2,321 psi; 1 L/min = 0.264172 US gpm; 1 kg = 2.20462 lb.

The Water Hydraulics Co. Janus pump catalogue V.01.25, pp. 2-4. Website product context checked 17 Oct 2026.

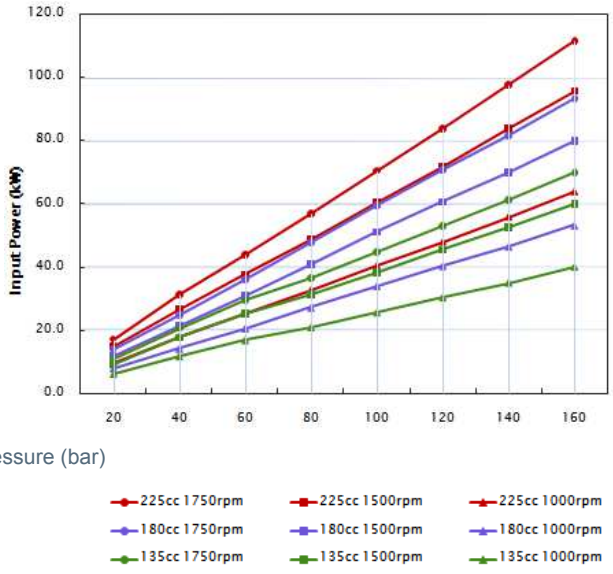
# P180 | Performance curves

Read displacement, speed and pressure together. Published curve data are preserved.

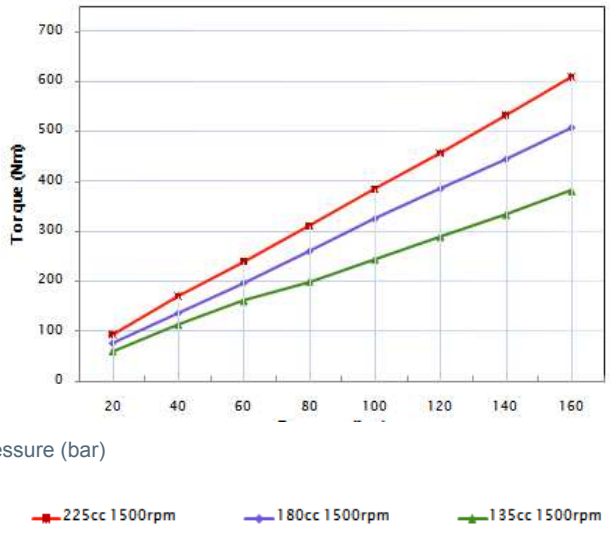
### Output flow (L/min)



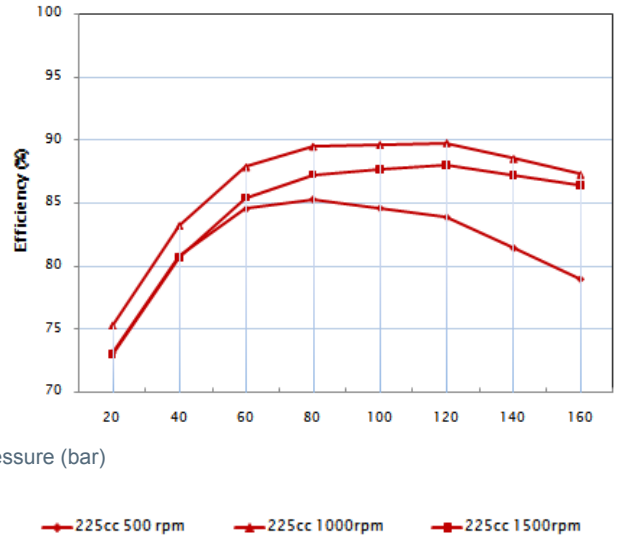

### Input power (kW)




### Input torque (N·m)




### Overall efficiency (%)




### Apply the curves to the duty

The range table lists a minimum of 135 cm³/rev; nominal ordering and some curve labels use 134. Match the quotation to its approved displacement and drawing.

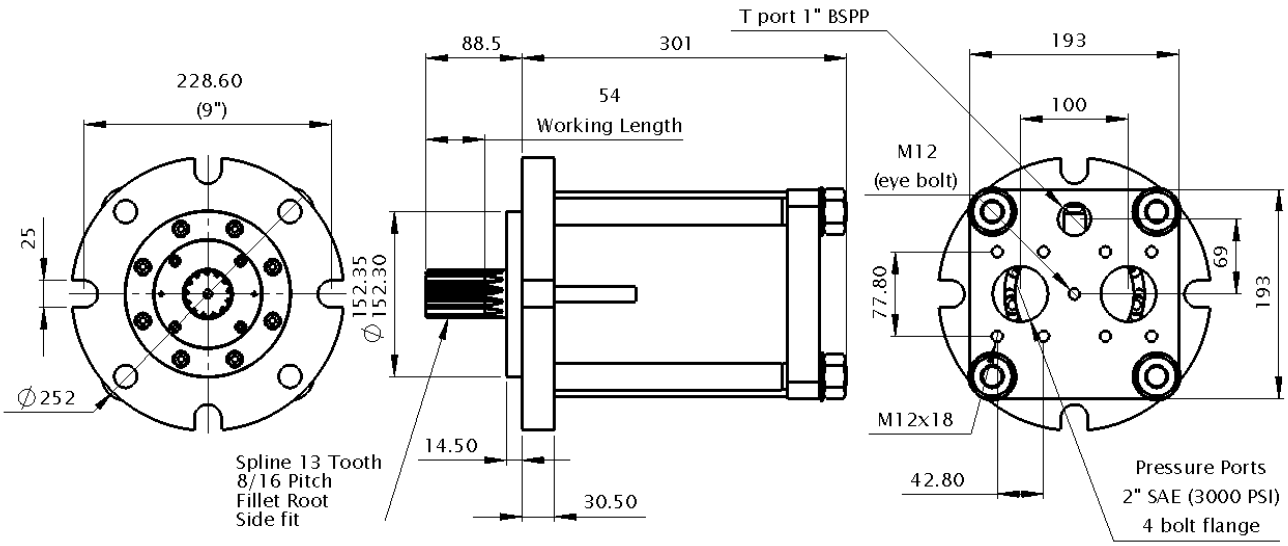
1 L/min = 0.264172 US gpm; 1 bar = 14.5038 psi; 1 kW = 1.34102 hp; 1 N·m = 0.737562 lbf·ft.

The Water Hydraulics Co. Janus pump catalogue V.01.25, p. 18.

# P180 | Dimensions & connections

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

### PB160-135W, PB160-180W & PB160-225W




Note:- The P180 pump is only supplied in the 3 port Boosted configuration with mechanical seal.

Port threads and flange classes are connection designations, not converted diameters or operating-pressure ratings. Select the exact configuration shown on the approved order drawing.

The Water Hydraulics Co. Janus pump catalogue V.01.25, p. 17.